Charbel Junior Nour

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Professional Summary

Highly motivated and detail-oriented Electrical Engineering student with 2 years of professional experience at Tesla and Moneris set to graduate from McMaster University in April 2025. Proficient in controls engineering, automation, systems optimization, and project management, with a strong passion for developing innovative solutions to complex challenges.

Skilled in leveraging industry-leading technologies, including TwinCAT3, MATLAB/Simulink, and various programming languages, to create efficient and impactful solutions. Excels in collaborative and cross-functional environments, contributing to the success of various large-scale projects in high-pressure settings.

Driven by a passion for controls engineering and a commitment to continuous improvement, I am eager to apply my expertise to tackle real-world challenges. Known for creativity, adaptability, and a results-oriented mindset, I bring a unique blend of technical knowledge, innovative thinking, and interpersonal skills to every project.

Skills

Technical Skills: Beckhoff TwinCAT3, PLC Programming, Control Systems, Structured Text, JavaScript, React.js, C++, Python, Java, Git, AWS, Arduino, RaspberryPI, MATLAB, Simulink, Autodesk Inventor, Jira, Microsoft 365
Special Interests: Guitar, Music Production, 3D Printing, Basketball, Skiing, Cycling, Running

Professional Experience

Tesla

Controls Engineer Intern — Jan. 2024 – May 2024

- Spearheaded the development of an **RFID**-based tool monitoring system that aims to **reduce** tooling maintenance issues by **over 90%**. Integrated **PLC logic** and started the development of a **custom HMI** to track service history and collect tooling lifespan data on **multiple Gigafactory assembly lines**.
- Supported the commissioning of 3 **automation** machines by leveraging **TwinCAT3 PLC** and **Safety PLC logic** for critical components, including **VFDs**, **IPCs**, various sensors, and roboticand pneumatic devices.
- Contributed to multiple **R&D** initiatives by testing with **time-of-flight lasers** and **profilometers** with an aim to **solve critical engineering challenges** to enhance machine performance in production environments.

Material Planner Intern — Sep. 2023 – Dec. 2023

- Developed and maintained 3 internal tools that streamlines the part reordering process through analysis of kitting history, conducting comprehensive **gap analyses**, and employing advanced **Excel VBA** techniques with existing company inventory – resulting in a significantly efficient database.
- Utilized these tools to build a software analysis team that focuses on the automation and standardization of many daily processes, resulting in a **reduction of duplicate shipments by 100%**.
- Facilitated international shipments, expedited missing parts, and streamlined kitting to **reduce technician downtime by 20%**, ensuring efficient workflows across Gigafactories and optimizing daily resource allocation.

Moneris

Technology Governance Intern — May 2023 – Aug. 2023

- Optimized the internal application database, **reducing its size by over 50%** through **cost center analysis**, a detailed **general ledger audit**, and advanced deduplication techniques within **Excel**. Developed **standard operating procedures (SOPs)** to maintain database efficiency and ensure consistent future updates.
- Completed data collection and validation by collaborating **cross-functionally** with teams and consulting with employees across departments, ensuring the database was accurate, comprehensive, and up-to-date.
- Enhanced internal compliance and governance practices by adhering the application inventory with corporate standards, improving transparency and accessability for future audits.

Software Engineer Intern — Jun. 2022 - Apr. 2023

- Developed and sustained an internal tool that increases productivity for **30+ employees**. The tool displays VISA, MasterCard, and AMEX transaction analytics by using **JavaScript** and **React** hooks to build a responsive and compatible UI that can manage dashboards, widgets, and layout customization.
- Improved code efficiency by **50%** by implementing **Redux** and a backend system hosted by **Parse** to reduce boilerplate, implement reducers, utilize dispatch, and **query user database** properties.
- Collaborated with the QA team in semi-weekly **agile** feedback sessions to identify and address platform issues, enhancing the reliability of a proprietary testing tool for payment platform evaluations.

Extracurriculars

McMaster EcoCar EV Challenge

Propulsion Controls and Modeling Team Member 2024 - Present

- **Modelling control systems** in **Simulink** for a Cadillac Lyriq EV through the use of documentation provided by **General Motors**, ensuring precise alignment with vehicle specifications and our team's performance goals.
- **Testing and validating control system requirements** using **Simulink**, ensuring compliance with benchmarks and functionality standards outlined by the competition organizers to achieve an **enhanced propulsion system**.
- Developing **energy-efficient** and **customer-focused control features** as part of a competition sponsored by the **US Department of Energy** and **General Motors** involving 13 universities across North America.

Projects

Autonomous Defense Drone System

- Developing an autonomous defense drone system to secure airspace by detecting, tracking, and intercepting unauthorized drones. The system integrates **multi-camera detection** and an autonomous drone to provide a solution to drone-related threats.
- Utilizing **RF Tx/Rx** modules to enable communication between the camera hub system and the drones. Implementing **YOLOv8**, an **AI camera**, and **control algorithms** in **Python** to command the motors and autonomously track, pursue, and intercept unauthorized drones.

Smart Curtain System

- Created a solution for sleep deprivation by designing an automated curtain. The app was coded in **React Native** and the backend is hosted on **AWS IoT** to store data for alarm functions and automated sunrise/sunset control.
- Implemented and deployed a **C++** program on an **Arduino Uno** to contact the backend server and control the stepper motor, enabling precise movement for rolling the curtain up and down automatically.

Autonomous RC Car

- Implemented self-driving algorithms coded in **C++ and Python** on an RC car to navigate through obstacles via **LiDAR** data processed through a **NVIDIA Jetson Nano** running on **Linux Ubuntu**.
- Coordinated **semi-weekly** agile team meetings to monitor progress, assign tasks, address challenges, and troubleshoot issues during development and testing phases.

Wireless Dog Doorbell System

- Designed and developed an **affordable wireless doorbell system** with an **Arduino Uno**, enabling dogs to alert owners when they need to go outside, facilitating effective potty training through dog-to-human communication.
- Built a mobile app using **React Native** to send real-time push notifications to owners, ultimately enhancing communication and improving the effectiveness of the training process.

Education

B. Eng., Electrical Engineering Expected Apr. 2025

McMaster University Hamilton, ON

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, High Performance Programming, Principles of Programming, Microprocessor Systems, Signals & Systems, Logic Design, Control Systems, Intro to Mechatronics, Power Electronics, Electric Motor Drives, Electromagnetics I & II, Electric Devices & Circuits
- **Extracurriculars:** McMaster EcoCar EV Challenge PCM Team Member, GDSC 2023 Incubator Challenge